Technical Writing with LaTeX
A Practical Demonstration

Ananya Siju
Department of Computer Science and Engineering
MNIT Jaipur

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Abstract

This document demonstrates essential LaTeX features for the Technical Writing laboratory: document structure, text formatting, lists, equations, tables, figures, cross-references, and bibliographic references.

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1 Introduction

LaTeX is a **document preparation system** especially useful for academic and technical writing. It supports *consistent formatting* and reproducible document generation. Visit the LaTeX Project for more information.

2 Document Structure

A LaTeX file begins with a document class and optional packages. The visible content belongs between the `\begin{document}` and `\end{document}` commands.

2.1 Advantages

- Consistent headings and page layout.
- High-quality mathematical notation.
- Automatic numbering and cross-references.
- Plain-text source compatible with Git.

2.2 Typical Workflow

1. Write a `.tex` source file.
2. Compile it using a LaTeX engine.
3. Review the PDF and correct any errors.
4. Commit the source to version control.

3 Mathematical Typesetting

Inline mathematics can express $E = mc^2$. Equation (1) shows the arithmetic mean of n observations.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{1}$$

4 Tables

Table 1 summarizes several documentation tools.

Tool	Main purpose	Typical output
LaTeX	Document typesetting	PDF
Beamer	Presentations	PDF slides
Doxygen	Code documentation	HTML or LaTeX
Git	Version control	Commit history

Table 1: Tools used in technical documentation

5 Figures

Figure 1 illustrates a simple documentation workflow. It is drawn directly in TikZ and does not require an external image file.



Figure 1: Basic LaTeX documentation workflow

6 Conclusion

LaTeX separates document content from much of its formatting and is well suited to structured technical writing. Additional information is available from the official LaTeX documentation [1].

References

[1] The LaTeX Project, *LaTeX – A document preparation system*. https://www.latex-project.org/.